COMP3670 / COMP6670: Introduction to Machine Learning
2026 Semester 2 Assignment 2

Instructions

Please read all instructions before you begin attempting the questions. Failure to follow the instructions may result in mark penalty.

- **Deadline:** 9 October 2026, 11:59pm (Canberra time)
- **Submission:**
  - Submit your solutions to all questions in **a single PDF file** via Gradescope.
  - Start each new question on a new page.
  - You are encouraged to typeset your solutions using LaTeX. If you opt for handwriting, make sure that it is clear and easy to read. Unreadable writing may not be graded.
- **Calculations:** Do not use a computer or calculator to assist with any calculations in this assignment. Neither will be available for calculations in the exam.
- **Grading:**
  - There are four questions. Total marks are 50.
  - After uploading your PDF file to Gradescope, you will be instructed to **assign page numbers to each question**. **A 10% penalty will be applied for each question you fail to assign page numbers to.** A simple guide for doing so is available here.
  - Show all your steps and proofs clearly. Correct numerical answers without supporting steps or proofs will receive no marks.
- **Academic Integrity:**
  - This is an **individual assignment**. You must work on your own.
  - The use of any generative AI tool is **not** allowed.
  - Your solutions are marked on your understanding and mathematical reasoning, not on the fluency of your English. Write clearly in your own words. Using generative AI to polish the wording of your solutions is prohibited and unnecessary.
  - For more detailed guidelines on academic integrity, see the Week 1 lecture slides and the ANU webpage on academic integrity.

Question 1: Rank and Nullity (14 marks)

Let $\mathbf{A} \in \mathbb{R}^{4\times 5}$ and $\mathbf{b}_t \in \mathbb{R}^4$ be given by

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 4 & 3 & 0 \\ 1 & -2 & -3 & 4 & -2 \\ 0 & 1 & 2 & -1 & 1 \\ 3 & -2 & -1 & 8 & -2 \end{pmatrix}, \qquad \mathbf{b}_t = \begin{pmatrix} 2 \\ -1 \\ 1 \\ t \end{pmatrix},$$

where $t \in \mathbb{R}$.

(a) Compute a row-echelon form of $\mathbf{A}$, showing your row operations. Give a basis for the column space of $\mathbf{A}$ using columns of the original matrix. [5 marks]

(b) Determine the rank and nullity of $\mathbf{A}$. Find a basis for its null space and verify the rank–nullity theorem. [5 marks]

(c) Determine all values of t for which $\mathbf{A}\mathbf{x} = \mathbf{b}_t$ is consistent. For each such value, describe all solutions $\mathbf{x} \in \mathbb{R}^5$. [4 marks]

Question 2: Orthogonal Projection (12 marks)

Let

$$\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 2 \\ -4 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \\ -6 \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} 3 \\ 2 \\ -4 \\ -1 \end{pmatrix},$$

and let $\mathcal{S} = \operatorname{span}\{\mathbf{a}, \mathbf{b}\} \subseteq \mathbb{R}^4$.

(a) Use Gram–Schmidt to construct an orthonormal basis of $\mathcal{S}$. [4 marks]

(b) Compute the orthogonal projection of $\mathbf{y}$ onto $\mathcal{S}$ and the corresponding residual. Verify that the residual is orthogonal to $\mathcal{S}$. [6 marks]

(c) Compute the distance from $\mathbf{y}$ to $\mathcal{S}$. [2 marks]

Question 3: Recommending Content on an Online Platform (17 marks)

An online platform recommends a short list of videos to each of n_u users from a catalogue of n_v videos, where both n_u and n_v are in the millions. It represents each user by an embedding $\mathbf{u} \in \mathbb{R}^{d_u}$ and each video by an embedding $\mathbf{v} \in \mathbb{R}^{d_v}$, where $d_u \neq d_v$ and both dimensions are between 100 and 200. The coordinates of the two embeddings have different meanings. As a data scientist working for the platform, your task is to design a system that improves the recommendations and, in turn, the platform's profit.

(a) Propose how clustering could be used in this recommendation system, and explain why it could help at this scale. [3 marks]

(b) Propose a matrix-based method for ranking videos by similarity to a user when their embeddings lie in different spaces. Specify the dimensions of your matrices, give a formula for the score of a user–video pair, and justify your choice of similarity measure. [3 marks]

(c) Suppose the platform also has users' viewing histories, including whether each viewed video received a like, a dislike, or no response. Formulate a supervised learning task to guide future recommendations for each user. Specify:

- the input features and target labels;
- the matrix parameters to be learned and their dimensions;
- the training loss as a function of these parameters.

Explain how the trained model would be used together with your clustering proposal from part (a) and your ranking method from part (b). After preprocessing, the time to produce a short recommendation list for one user must be independent of n_v; approximate recommendations are acceptable. Explain how your approach meets this requirement and how its memory use scales with n_u, n_v, d_u, d_v and any other quantities you introduce. State the assumptions behind your bounds. [9 marks]

(d) Identify one potential drawback of your approach in part (c) and explain how it could affect the recommendations. [2 marks]

Question 4: Matrix Products and Rank (7 marks)

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be an arbitrary matrix.

(a) Prove that $\text{rank}(\mathbf{A}^\mathsf{T}\mathbf{A}) = \text{rank}(\mathbf{A})$. [5 marks]

(b) Deduce that $\mathbf{A}^\mathsf{T}\mathbf{A}$ is invertible if and only if the columns of $\mathbf{A}$ are linearly independent. [2 marks]